

Large Scale Data Management Assignment 1

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1. Part 1 :

The file used for the first part of the assignment is the txt file of the Gothic horror novel 'Dracula', provided by the URL <https://www.gutenberg.org/files/345/345-0.txt>.

The steps used to run the WordCount MapReduce job are the following:

- The command `wget` was used with the URL provided above in order to download the .txt file of the novel *Dracula* from Project Gutenberg.
- The command `docker cp Dracula.txt namenode:/` was used to copy the file from our Virtual Machine to the namenode container. This step is necessary because our local machine (the VM we created) cannot directly interact with HDFS inside the Docker container. Therefore, the file must first be copied into the namenode and then inserted into HDFS.
- The file was placed into the HDFS input folder using the command:
`docker exec namenode hdfs dfs -put -f /input.txt /user/hdfs/input/input.txt`
- A modification was made to the Driver.java file so that the application executes using the *Dracula.txt* file instead of the default input file.
- Due to the modification in the Driver file, a new executable .jar file had to be created using the command `mvn clean install`. The newly generated jar file was then copied to the namenode container in order to execute the MapReduce job.

The log of the execution of the application, along with the full output file produced by the WordCount program, are included in the .zip file provided for the assignment.

2. Part 2:

The steps used to run the Spotify MapReduce job are the following:

- The Spotify dataset (spotify.csv) was copied into the namenode container using `docker cp`, and then inserted into HDFS under `/user/hdfs/input/spotify.csv` using `docker exec namenode hdfs dfs -put`.
- The header line of the CSV file was ignored in the Mapper, as required.
- Each line of the CSV was parsed using a CSV-safe split regex (to handle commas inside quotes).
- The Mapper emits key-value pairs where the key is `country:YYYY-MM` (derived from country and the month of `snapshot_date`), and the value contains the song name and its danceability.
- The Reducer computes (i) the most danceable song (maximum danceability) and (ii) the average danceability for each country-month key.
- The project was rebuilt using `mvn clean install` and the updated jar was executed in Hadoop using `hadoop jar`.

The execution log and the full output file are included in the submitted zip as `SpotifyDanceabilityTerminalLog.txt` and `SpotifyDanceabilityOutput.txt`.